Generate Word Search Puzzle:

```latex
\begin{algorithm}
\caption{Generate Word Search Puzzle}
\begin{algorithmic}[1]

\State \textbf{Function} generateGrid(wordList, gridSize)
\State \hspace{0.5cm} size $\leftarrow$ gridSize
\State \hspace{0.5cm} Create an empty 2D matrix[size][size]

\State \hspace{0.5cm} \For{each cell in the matrix}
\State \hspace{1cm} Fill the cell with a placeholder (like '.')
\State \hspace{0.5cm} \EndFor

\State \hspace{0.5cm} \For{each word in wordList}
\State \hspace{1cm} placed $\leftarrow$ false

\State \hspace{1cm} \While{placed = false}
\State \hspace{1.5cm} Pick a random direction (horizontal, vertical, or diagonal)
\State \hspace{1.5cm} Randomly choose a starting (x, y) position

\State \hspace{1.5cm} \If{the word actually fits in that direction}
\State \hspace{2cm} placeWord(word, startX, startY, direction)
\State \hspace{2cm} placed $\leftarrow$ true
\State \hspace{1.5cm} \EndIf
\State \hspace{1cm} \EndWhile

\State \hspace{0.5cm} \EndFor

\State \hspace{0.5cm} \For{each remaining empty cell}
\State \hspace{1cm} Replace placeholder with a random A–Z letter
\State \hspace{0.5cm} \EndFor

\State \hspace{0.5cm} \Return matrix
\State \textbf{End Function}

\end{algorithmic}
\end{algorithm}
```

Place Word:

```
\begin{algorithm}
\caption{Place a Word in the Grid}
\begin{algorithmic}[1]

\State \textbf{Function} placeWord(word, x, y, direction)

\State \hspace{0.5cm} \For{$i \leftarrow 0$ to length(word) - 1}

\State \hspace{1cm} \If{direction = "horizontal"}
\State \hspace{1.5cm} matrix[y][x + i] $\leftarrow$ word[i]

\State \hspace{1cm} \ElsIf{direction = "vertical"}
\State \hspace{1.5cm} matrix[y + i][x] $\leftarrow$ word[i]

\State \hspace{1cm} \ElsIf{direction = "diagonal"}
\State \hspace{1.5cm} matrix[y + i][x + i] $\leftarrow$ word[i]

\State \hspace{1cm} \EndIf

\State \hspace{0.5cm} \EndFor

\State \textbf{End Function}

\end{algorithmic}
\end{algorithm}
```

Word Checking:

```
\begin{algorithm}
\caption{Check User-Selected Word Coordinates}
\begin{algorithmic}[1]

\State \textbf{Function} checkWord(startX, startY, endX, endY, wordList, grid)

\State \hspace{0.5cm} selectedWord $\leftarrow$ ""
\State \hspace{0.5cm} dx $\leftarrow$ sign(endX - startX)
\State \hspace{0.5cm} dy $\leftarrow$ sign(endY - startY)

\State \hspace{0.5cm} x $\leftarrow$ startX
\State \hspace{0.5cm} y $\leftarrow$ startY

\State \hspace{0.5cm} \While{(x, y) hasn't reached (endX, endY)}
\State \hspace{1cm} selectedWord += grid[y][x]
\State \hspace{1cm} x $\leftarrow$ x + dx
\State \hspace{1cm} y $\leftarrow$ y + dy
\State \hspace{0.5cm} \EndWhile

\State \hspace{0.5cm} selectedWord += grid[endY][endX]

\State \hspace{0.5cm} \If{selectedWord is in wordList}
\State \hspace{1cm} \Return true
\State \hspace{0.5cm} \Else
\State \hspace{1cm} \Return false
\State \hspace{0.5cm} \EndIf

\State \textbf{End Function}

\end{algorithmic}
\end{algorithm}
```